CYBERSECURITY LABS

• KALI LINUX BASICS :

COMMANDS PRACTICED

> ls - list files
> cd - change directory
> pwd - show current directory
> chmod - change permissions
> grep -
> ps aux - show running processes
> top - live process monitor (to stop press q)
> mkdir - create new directory

SKILLS LEARNED

> FILE PERMISSIONS
> PROCESS MONITORING
> LOG SEARCHING

• NMAP SCANNING :

/SCANS

COMMANDS PRACTICED

> nmap -Pn scanme.nmap.org/
BASIC SCAN

> nmap -sV -Pn scanme.nmap.org/
VERSION DETECTION

> nmap -A -Pn scanme.nmap.org/
Aggressive scan

SKILLS LEARNED

> PORT SCANNING
> SERVICE IDENTIFICATION
> OS DETECTION.

• WIRESHARK ANALYSIS :

COMMAND/TRAFFIC ANALYZED

> Wireshark - SELECT : eth0 thn START
> http://example.com - IN FIREFOX BROWER
> HTTP TRAFFIC → GET REQUEST : GET/HTTP/1.1
> DNS TRAFFIC
> TCP TRAFFIC → (TRANSMISSION CONTROL PROTOCOL)
→ • SOURCE IP
• DESTINATION IP

IN WIRESHARK FILTER BAR

DNS QUERY : STANDARD QUERY
A EXAMPLE.COM

SKILLS LEARNED

> PACKET CAPTURE
> PROTOCOL FILTERING
> IP ADDRESS IDENTIFICATION.

TRYHACKME ✔ PRE SECURITY (LEGACY)

• INTRODUCTION TO CYBERSECURITY:

> OFFENSIVE (RED TEAM) VS DEFENSIVE SECURITY (BLUE TEAM)

RED TEAM

- Focus: Find weakness by attacking ethically
- Perform penetration testing
- Exploit vulnerabilities (legally)
- Use tools like Kali Linux, Metasploit, Nmap
- Report vulnerabilities to improve security

BLUE TEAM

- Focus: Protect systems from attacks
- Monitors networks and logs
- Install patches and updates
- Use firewalls, IDS, antivirus
- Respond to incidents.

> Practiced finding and accessing a hidden admin panel in a web application.

> Learned how to manipulate transactions to make an account balance positive.

• NETWORK FUNDEMENTALS:

> NETWORK is the key term for devices that are connected together.

> THE INTERNET is one gaint networks that consist of many, many small networks within itself. TIM BERNERS-LEE invented the World Wide Web.

> Each section of IP address is called OCTET.

> Media Access Control (MAC)

> Ping is a network tool. Ping uses ICMP (Internet Control Message Protocol).

> Address Resolution Protocol (ARP)

> MAC ADDRESS is used as a physical identifier for a device on a network

> IP ADDRESS is used as a logical identifier for a device on a network.

> Dynamic Host Configuration Protocol (DHCP)

> Open Systems Interconnection (OSI)

7. APPLICATION
6. PRESENTATION
5. SESSION
4. TRANSPORT
3. NETWORK
2. DATA LINK
1. PHYSICAL

> Network Interface Card (NIC)

> Open Shortest Path First (OSPF)

> Routing Information Protocol (RIP)

> Transmission Control Protocol (TCP)

> User Datagram Protocol (UDP)

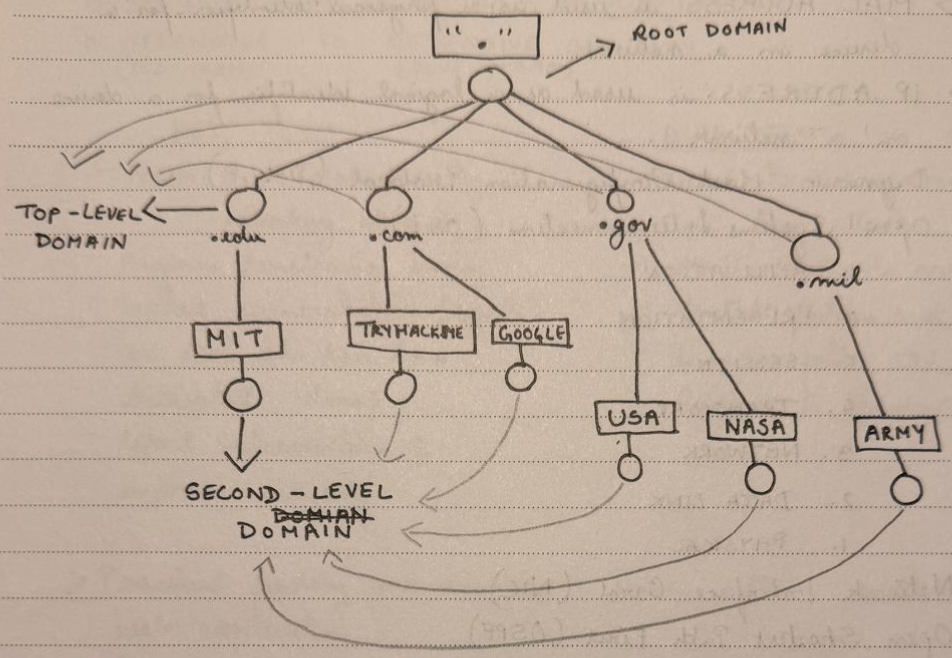
> Graphical User Interface (GUI), Domain Name System (DNS)

> ROUTER is the name of the device that is used to configure port forwarding.

- HOW THE WEB WORKS:

> Domain Name System (DNS) provides a simple way for us to communicate with devices on the Internet without remembering complex numbers.

> DOMAIN HIERARCHY



> 63 is the maximum length of a subdomain
> 253 is the maximum length of a domain name.
> There are two types of TLD, gTLD (Generic Top level) and ccTLD (Country Code Top level Domain).
> HTTP stands for HYPER TEXT TRANSFER PROTOCOL.
> HTTPS stands for HYPER TEXT TRANSFER PROTOCOL SECURE.

Scheme — Host domain — Path — Fragement

http://user:password@tryhackme.com:80/view-room?id=1#task3

User — Port — Query string

- **LINUX FUNDAMENTALS:**
  - > 1991 was the first release of a Linux operating system.
  - > ls = listing
    - Command = Full Name
    - cd = change Directory
    - cat = concatenate
    - pwd = print working directory